3.1.2 Amount of substance: Sheet 7

Required practical 1: method and apparatus · 19 marks · about 25 minutes

Key information

- **RP1:** make up a volumetric solution and carry out a simple acid–base titration.
- Weigh the solid in a weighing bottle, tip into a beaker, then reweigh the empty bottle (weighing by difference).
- Dissolve in less than 250 cm^3 of distilled water, transfer to the volumetric flask with all washings, make up so the bottom of the meniscus is on the line, then invert to mix.
- Rinse the burette and pipette with the solution they will hold; rinse the conical flask with distilled water only.
- Remove the funnel and make sure the jet below the tap is filled before taking the first burette reading.

Section A: Self-quiz

01 A student rinses the burette before filling it with sodium hydroxide solution. State why the student should use sodium hydroxide solution rather than water for the final rinse of the burette.

[1 mark]

02 Before the first titration, a 25 cm^3 pipette is rinsed with a small volume of the 0.100 mol dm^{-3} barium hydroxide solution it will be used to measure. State why it is good practice to rinse the pipette in this way.

[1 mark]

03 25.0 cm^3 samples of NaOH solution were taken by pipette from a beaker. These were then titrated with an aqueous solution of ethanoic acid. The concentration of ethanoic acid calculated from the experimental results was found to be lower than the actual value. Which of these could explain the difference?

[1 mark]

A Rinsing the pipette with distilled water before filling with NaOH
B Rinsing the burette with distilled water before filling with ethanoic acid
C Rinsing the walls of the conical flask with distilled water during the titration
D Rinsing the beaker with distilled water before filling with NaOH

04 A student rinsed the apparatus before starting an acid–base titration. The results of the titration showed that the volume of acid added from the burette was larger than expected. Which is a possible reason for this?

[1 mark]

A The conical flask was rinsed with water before the titration.
B The walls of the conical flask were rinsed with water during the titration.
C The pipette was rinsed only with water.
D The burette was rinsed only with water.

05 A burette has been filled with solution using a funnel. Give two practical steps needed before recording the initial burette reading.

[2 marks]

Section B: Exam questions

06 Propanedioic acid is a solid diprotic acid (M_r = 104.0).

06.1 Describe how to prepare 250 cm^3 of an aqueous standard solution of propanedioic acid containing an accurately measured mass of the acid.

Include essential practical details in your answer.

[6 marks]

06.2 Calculate the mass, in mg, of propanedioic acid needed to prepare 250 cm^3 of a 0.00500 mol dm^{-3} solution.

[2 marks]

07 Ethanedioic acid ($H_2C_2O_4$) is a diprotic acid. A student carried out a titration with sodium hydroxide solution to determine the mass of the acid in a solution. The student repeated the titration until concordant titres were obtained.

$H_2C_2O_4(aq) + 2NaOH(aq) \rightarrow Na_2C_2O_4(aq) + 2H_2O(l)$

07.1 The student found that 25.0 cm^3 of the ethanedioic acid solution reacted completely with 25.30 cm^3 of 0.500 mol dm^{-3} sodium hydroxide solution.

Calculate the mass, in mg, of the acid in 25.0 cm^3 of this solution.

[4 marks]

07.2 The student used a wash bottle containing deionised water when approaching the end point to rinse the inside of the conical flask.

Explain why this improved the accuracy of the titration.

[1 mark]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	Use of water would dilute the NaOH / change the concentration of NaOH / to ensure the concentration of the NaOH is not changed	Source: 7404/2 Jun 2018 Q02.1 Ignore reference to weakening the solution, watering down the solution, contaminate Allow it would give a titre value that is larger	1
02	So that the titration is done with known concentration of $Ba(OH)_2$	Source: 7405/1 Jun 2023 Q06.3 Allow so that water does not dilute the solution Allow remove water / prevent contamination	1
03	**B**	Source: 7404/1 Jun 2017 Q16	1
04	**D**	Source: 7405/3 Nov 2020 Q36	1
05	Remove funnel Ensure jet is filled / no (air) bubbles	Source: 7405/3 Nov 2021 Q01.7 (figure removed) Allow open tap to fill space below tap	2
06.1	M1 weigh sample in bottle / boat / container M2 transfer to (conical) flask / beaker (or suitable container) and wash all sample in or re-weigh bottle / boat / container M3 dissolve sample in (deionised / distilled) water (if volume of water is specified, must be less than 250 cm^3) M4 add into volumetric flask with washings M5 make up to mark / 250 cm^3 in volumetric flask M6 shake / invert (to give a homogeneous solution; must be after made up to mark)	Source: 7404/2 Jun 2019 Q07.2–07.3 Max 4 marks for candidates who add any substance other than water Penalise M1 for weighing out wrong substance or using the acid as a liquid or solution Allow graduated flask for volumetric flask Candidates may dissolve sample directly in volumetric flask: M1 weigh; M2 add into volumetric flask; M3 wash all sample in or re-weigh; M4 dissolve in water; M5 make up to mark; M6 shake / invert	6
06.2	M1 moles of acid = 0.00500 × 250 ÷ 1000 (= 0.00125) M2 mass of acid = 0.00125 × 104(.0) = 0.130 g = 130 (mg)	Final answer must be at least 2 sf 0.13(0) scores 1 mark	2
07.1	M1 Amount NaOH = 0.02530 × 0.500 = 0.01265 mol M2 Amount acid = 0.006325 mol (ie M1 ÷ 2) M3 M_r = 90(.0) M4 mass acid = 569 (mg) (allow 567 to 576)	Source: 7404/2 Jun 2017 Q05.1–05.2 Allow ECF at each stage 0.567 – 0.590 = 3 marks 1134 – 1180 = 3 marks (moles of NaOH not divided by 2)	4
07.2	Ensures all reactants / acid / alkali are in the mixture / reaction OR the idea that some of the acid / alkali would be on the sides of the flask	The idea that it is the transfer of all the acid / alkali alone is not enough	1